EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Which of the following statements is true?

(A) $3 \subseteq \{1, 3, 5\}$ (B) $3 \in \{1, 3, 5\}$ (C) $\{3\} \in \{1, 3, 5\}$ (D) $\{3, 5\} \in \{1, 3, 5\}$

MSN001

2. How many of the following statement is/are correct

P : In roster form, the order in which the elements are listed is immaterial

Q : While writing a set in roster form, an element is not generally repeated.

R : The collection of five most renowned mathematicians of the world is not a set.

S : A collection of most dangerous animals of the world is a set

(A) 1 (B) 2 (C) 3 (D) 4

MSN012

3. How many of the following statement is/are sets.

(i) The collection of all the months of a year beginning with the letter J.

(ii) The collection of ten most talented writers of India.

(iii) A team of eleven best-cricket batsmen of the world.

(iv) The collection of all boys in your class.

(v) The collection of all-natural numbers less than 100.

(A) 1 (B) 2 (C) 3 (D) 4

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MSN013

4. How many of the following statement is/are correct?

(a) Let $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Then A is the empty set, because there is no natural number between 1 and 2.

(b) $B = \{x : x^2 - 2 = 0 \text{ and } x \text{ is rational number}\}$. Then B is the empty set because the equation $x^2 - 2 = 0$ is not satisfied by any rational value of x.

(c) $C = \{x : x \text{ is an even prime number greater than 2}\}$. Then C is the empty set, because 2 is the only even prime number

(d) $D = \{x : x^2 = 4, x \text{ is odd}\}$. Then D is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of x.

(A) 1 (B) 2 (C) 3 (D) 4

MSN014

5. How many of the following statement is/are correct ?

(a) Let W be the set of the days of the week. Then W is finite.

(b) Let S be the set of solutions of the equation $x^2 - 16 = 0$. Then S is finite.

(c) Let G be the set of points on a line. Then G is infinite.

(A) 0 (B) 1 (C) 2 (D) 3

MSN015

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6. How many of the following statement is/are correct?

P. $A \subset B$ if $a \in A \Rightarrow a \in B$

Q. A is a subset of B if a is an element of A implies that a is also an element of B

R. ϕ is a subset of every set

S. Every set A is a subset of itself

(A) 4 (B) 1 (C) 2 (D) 3

MSN016

7. Which of the following is a null set?

(A) $A = \{x : x > 1 \text{ and } x < 1]$ (B) $B = \{x : x + 3 = 3\}$

(C) $C = \{\phi\}$ (D) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$

MSN002

8. The set of intelligent students in a class is -

(A) a null set (B) a singleton set

(C) a finite set (D) not a well defined collection

MSN102

9. The set $A = \{x : x \in R, x^2 = 16 \text{ and } 2x = 6\}$ is

(A) Null set (B) Singleton set

(C) Infinite set (D) not a well defined collection

MSN053

10. Let $S = \{1,2\}$, number of elements in the set $\{(A, B) : A \cup B = S\}$

(A) 4 (B) 6 (C) 9 (D) 10

MSN010

11. Let A and B be two sets in the universal set. Then $A - B$ equals-

(A) $A \cap B'$ (B) $A' \cap B$ (C) $A \cap B$ (D) none of these

MSN003

12. If $A \subseteq B$, then $A \cap B$ is equal to-

(A) A (B) B (C) A' (D) B'

MSN004

13. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-

(A) A (B) B (C) A' (D) B'

MSN005

14. If A and B are not disjoint, then $n(A \cup B)$ is equal to-

(A) $n(A) + n(B)$ (B) $n(A) + n(B) - n(A \cap B)$

(C) $n(A) + n(B) + n(A \cap B)$ (D) $n(A).n(B)$

MSN006

15. If the number of elements in $(A - B) - C$, $(B - C) - A$, $(C - A) - B$ and $A \cap B \cap C$ is 10, 15, 20, and 5 respectively then the number of elements in $(A \Delta B) \Delta C$ is

(A) 35 (B) 50 (C) 40 (D) 45

MSN101

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16. Number of integers in between 200 to 2023 (both 200 and 2023 including) are divisible by 2 or 5

(A) 1093 (B) 1094 (C) 1275 (D) 1277

MSN008

17. Number of integers in between 500 to 2023 are divisible by 3 or 5 but not multiple of 2

(A) 356 (B) 355 (C) 710 (D) 711

MSN009

18. If $x + y = 1$ and $x^2 + y^2 = 2$ then the value of $x^4 + y^4$ equals

(A) 7 (B) 6 (C) $\frac{7}{2}$ (D) $\frac{19}{4}$

MSN017

19. Pranshi has written down one integer two times and another integer three times. The sum of the five numbers is 100, and one of the numbers is 28. The other number is

(A) 6 (B) 8 (C) 10 (D) 12

MSN018

20. If $x, y \in \mathbb{N}$ and $xy - 3x - 2y = 54$, then number of ordered pairs (x, y) is

(A) 7 (B) 8 (C) 12 (D) 24

MSN019

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Which of the following are correct?
(A) $[a, b)$ is an interval from a to b, including a but excluding b.
(B) $(a, b]$ is an interval from a to b, including b but excluding a.
(C) $[a, b)$ is an interval from a to b, including b but excluding a.
(D) $(a, b]$ is an interval from a to b, including a but excluding b.

MSN033

2. In which of the following options, $A \neq B$:
(A) $A = \{x : x + 2 = 3\}$, $B = \{x : x \in \mathbb{N}$ and is less than $2\}$
(B) $A = \{x : x \in \mathbb{N}$ and $3x - 1 < 2\}$, $B = \{x : x \in \mathbb{W}$ and $3x - 1 < 2\}$
(C) $A = \{x : x \in \mathbb{N}$ and is prime factor of $36\}$, $B = \{1,2,3,4,6,9,12\}$
(D) $A = \{x : x \in I$ and $x^2 \leq 4\}$, $B = \{x : x \in \mathbb{R}$ and $x^2 - 3x + 2 = 0\}$

MSN036

3. Which of the following are examples of the null set
(A) Set of odd natural numbers divisible by 2
(B) Set of even prime numbers
(C) $\{x : x$ is a natural number $x < 5$ and $x > 7\}$
(D) $\{y : y$ is a point common to any two distinct parallel lines$\}$

MSN037

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4. Consider the set ϕ, $A = \{1,3\}$, $B = \{1,5,9\}$, $C = \{1,3,5,7,9\}$. Which of the following statements are correct?
(A) $\phi \subset B$ as ϕ is a subset of every set.
(B) $A \not\subset B$ as $3 \in A$ and $3 \notin B$
(C) $A \subset C$ as $1,3 \in A$ also belongs to C
(D) $B \subset C$ as each element of B is also an element of C.

MSN038

5. Which of the following options are incorrect
(A) $\{2,3,4,5\}$ and $\{3,6\}$ are disjoint sets
(B) $\{a, e, i, o, u\}$ and $\{a, b, c, d\}$ are disjoint sets
(C) $\{2,6,10,14\}$ and $\{3,7,11,15\}$ are disjoint sets
(D) $\{2,6,10\}$ and $\{3,7,11\}$ are disjoint sets

MSN039

6. If $U = \{1,2,3,4,5,6,7,8\}$, $A = \{1,2,3,5,6\}$ and $B = \{2,3,4,7,8\}$ which of the following are correct
(A) $A \cup B = U$
(B) $B - A = \{1,5,6\}$
(C) $A' \cup B = \{2,3,4,7,8\}$
(D) $(A - B)' = \{2,3,4,7,8\}$

MSN030

7. Which of the following are correct?
(A) $[2, 10] - \{2, 10\} = (2, 10)$
(B) $[3, 5] - \{5\} = (3,5]$
(C) $(-3, 5] - \{-3\} = (3, 5]$
(D) $[-2, 15] - (2, 15] = [-2, 2]$

MSN031

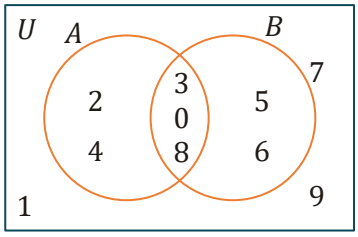
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8. If $A \subseteq B$ then which of the following option(s) is/are correct?

(A) $A' \subseteq B'$ (B) $B' \subseteq A'$ (C) $A \cap B' = \phi$ (D) $A' \cap B = \phi$

MSN032

9. From the adjoining Venn-diagram, find which of the following are correct :



(A) $A' = \{1,5,6,7,9\}$ (B) $B' = \{1,2,4,7,9\}$

(C) $(A \cap B)' = \{1,2,4,5,6,7,9\}$ (D) $(A - B)' = \{1,2,4,5,6,7,9\}$

MSN034

10. Which of the following option(s) is/are true?

(A) If $A \subset B$ and $A \neq B$, then A is called a proper subset of B

(B) If $A \subset B$, then B is called superset of A.

(C) $A = \{1,2,3\}$ is NOT a proper subset of $B = \{1,2,3,4\}$

(D) If a set A has only one element, we call it a singleton set.

MSN035

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EXERCISE (S)

1. Sets A and B have 3 and 6 elements respectively. What can be the sum of minimum and maximum number of elements in $A \cap B$?

MSN109

2. A class has 175 students. The following data shows the number of students obtaining one or more subjects: Mathematics 100, Physics 70, Chemistry 40, Mathematics and Physics 30, Mathematics and Chemistry 28, Physics and Chemistry 23, Mathematics & Physics & Chemistry 18. How many students have offered Mathematics alone?

MSN057

3. Of the members of three athletic teams in a school 21 are in the cricket team, 26 are in the hockey team and 29 are in the football team. Among them, 14 play hockey and cricket, 15 play hockey and football, and 12 play football and cricket. Eight play all the three games. The total number of members in the three athletic teams is :

MSN054

4. In a town of 10,000 families it was found that 40% families buy newspaper A, 20% families buy newspaper B and 10% families buy newspaper C, 5% families buy A and B, 3 % buy B and C and 4% buy A and C. If 2% families buy all the three news papers, then find number of families which buy newspaper A only.

MSN056

5. In DDPS – ALLEN, out of 400 students, 300 students opt for at least one subject out of Biology, Chemistry and Mathematics. If 120 students opted for Biology, 150 students opted for Chemistry, 200 students opted for Mathematics and 30 opted for all the three subjects, find the number of students who opted for exactly two subjects.

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MSN024

6. Let $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then find $n(A' \cap B')$

MSN104

7. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples, number of integral values of x is.

MSN022

8. How many ordered pairs of integers which satisfy the equation $\frac{1}{m}+\frac{2}{n}=\frac{1}{2}$?

MSN023

9. In a Zoo, there are 6 Bengal white tigers and 6 Bengal royal tigers. Out of these tigers, 5 are males and 10 are either Bengal royal tigers or males. Find the number of female Bengal white tigers in the Zoo.

MSN025

10. In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. Find the number of newspaper.

MSN055

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EXERCISE - JEE (Main) PYQ

1. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :- **[AIEEE 2009]**

(1) $B = C$ (2) $A \cap B = \phi$ (3) $A = B$ (4) $A = C$

MSN026

2. In a class of 140 students numbered 1 to 140, all even numbered students opted mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is : **[JEE (Main) 2019]**

(1) 102 (2) 42 (3) 1 (4) 38

MSN027

3. Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :- **[JEE (Main) 2019]**

(1) 12.8 (2) 13.5 (3) 13.9 (4) 13

MSN028

4. Let $\mathbb{Z}$ be the set of integers. If $A=\left\{x\in\mathbb{Z}:2^{(x+2)\left(x^2-5x+6\right)}=1\right\}$ and $B=\left\{x\in\mathbb{Z}:-3<2x-1<9\right\}$, then the number of subsets of the set $A \times B$, is: **[JEE (Main) 2019]**

(1) 2^{18} (2) 2^{10} (3) 2^{15} (4) 2^{12}

MSN029

5. Let $X = \{n \in N : 1 \le n \le 50\}$. If $A = \{n \in X : n$ is a multiple of $2\}$ and $B = \{n \in X : n$ is a multiple of $7\}$, then the number of elements in the smallest subset of X containing both A and B is____

[JEE (Main) 2020]

MSN049

6. A survey shows that 73% of the persons working in an office like coffee, whereas 65% like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be:

[JEE (Main) 2020]

(1) 63 (2) 38 (3) 54 (4) 36

MSN050

7. Let $A = \{1,2,3,4,5,6,7\}$ and $B = \{3,6,7,9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is_____ **[JEE (Main) 2022]**

MSN051

8. An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events? **[JEE (Main) 2023]**

(1) 10 (2) 9 (3) 21 (4) 15

MSN052

Rankers Academy JEE

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9. In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the most number of students who studied all the three subjects. Then $m + n$ is equal to ____ **[JEE (Main) 2024]**

MSN105

10. Let $A = \{n \in [100, 700] \cap N : n$ is neither a multiple of 3 nor a multiple of 4$\}$. Then the number of elements in A is **[JEE (Main) 2024]**

(1) 300 (2) 280 (3) 310 (4) 290

MSN106

11. A group of 40 students appeared in an examination of 3 subjects – Mathematics, Physics & Chemistry. It was found that all students passed in at least one of the subjects, 20 students passed in Mathematics, 25 students passed in Physics, 16 students passed in Chemistry, at most 11 students passed in both Mathematics and Physics, at most 15 students passed in both Physics and Chemistry, at most 15 students passed in both Mathematics and Chemistry. The maximum number of students passed in all the three subjects is _____. **[JEE (Main) 2024]**

MSN107

12. Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m, n)$ from the point $Q(-2, -3)$ is **[JEE (Main) 2024]**

(1) 10 (2) 6 (3) 4 (4) 8

MSN108

Rankers Academy JEE

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EXERCISE - JEE (Advanced) PYQ

1. If X and Y are two sets, then $X \cap (X \cup Y)^c$ equals

(A) X (B) Y (C) ϕ (D) none of these

MSN045

2. If $A = \{1, 2, 3, 4, 5\}$, then the number of proper subsets of A is-

(A) 120 (B) 30 (C) 31 (D) 32

MSN046

3. Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$?

(A) 3 (B) 6 (C) 9 (D) 18

MSN048

Rankers Academy JEE

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	D	D	A	A	D	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	B	B	B	A	C	B	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C,D	A,C,D	A,B,C,D	A,B	A,C,D	A,D	B,C	A,B,C	A,B,D

EXERCISE (S)

1.	3	**2.**	60	**3.**	43	**4.**	3300	**5.**	110
6.	300	**7.**	25	**8.**	7	**9.**	2	**10.**	25

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	3	3	29	4	112	3	45	1
Que.	11	12								
Ans.	10	1								

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3
Ans.	C	C	B

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